



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Build a Human-AI Workforce with Autonomous AI Agents

TGS Ref No: TGS-2024043854

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	12 July 2026	Initial release — 3-day lesson plan for the autonomous-AI-agent labs (Hermes Agent, OpenClaw, Paperclip).	Dr. Alfred Ang
1.1	14 July 2026	Revised to a 2-day plan (Day 1: Topics 1 & 2; Day 2: Topic 3 & final assessment). Hermes expanded to 14 labs, OpenClaw to 12, Paperclip to 8 (34 labs total). Learning outcomes aligned to the Course Proposal / TSC; per-lab reference videos added.	Dr. Alfred Ang

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Course Information

Course Title	Build a Human-AI Workforce with Autonomous AI Agents
WSQ Course Reference	TGS-2024043854
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on autonomous-AI-agent labs (local machine + Docker)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyse AI (LLM-based) applications across a range of industries to identify their capabilities and limitations.
- LO2: Establish the relationship between AI algorithm/agent design and chatbot/agent efficiency.
- LO3: Evaluate and improve the effectiveness of AI (RAG and agent) applications in product development.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on agent-building tasks that follow the course labs, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Hermes Agent (Topic 1) & OpenClaw (Topic 2)

Time	Duration	Topic / Activity
9:30–9:50	20 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM)
9:50–10:30	40 min	Topic 1 — Hermes Agent — Your Autonomous Personal Chief-of-Staff: what an autonomous AI agent is, the agent stack (model, memory, skills, tools) and human-in-the-loop governance

		(concepts + demo)
10:30–11:00	30 min	Hands-on: Lab 1: Install and Setup Hermes; Lab 2: Deployment — Local Install & Hermes Desktop App; Lab 3: Memory and Plugins
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Hands-on: Lab 4: Skills; Lab 5: Providers & Model; Lab 6: MCP and Tools; Lab 7: Cron Jobs & Automation; Lab 8: Subagents & Delegation; Lab 9: Profile and Kanban
13:00–14:00	60 min	Lunch break
14:00–14:40	40 min	Hands-on: Lab 10: Security; Lab 11: Use Case — Make a Video with Hyperframe; Lab 12: Visualization; Lab 13: Build a Multi-Agent Workflow; Lab 14: Webhook
14:40–15:20	40 min	Topic 2 — OpenClaw — Automating a Business Back-Office: the OpenClaw gateway daemon, providers, channels, skills, tools, dreaming and crons for a business back-office (concepts + demo)
15:20–15:35	15 min	Tea break
15:35–18:15	160 min	Hands-on: Lab 15: Install OpenClaw; Lab 16: Models & Providers; Lab 17: Channels; Lab 18: Skills; Lab 19: Tools & Integrations; Lab 20: Commands; Lab 21: Cron Jobs & Heartbeat; Lab 22: Memory & Context; Lab 23: Security; Lab 24: Multi-Agent Workforce; Lab 25: Dreaming — Idle Reflection & Self-Improvement; Lab 26: Use Cases & Key Functions
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Paperclip (Topic 3) & Final Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–10:30	45 min	Topic 3 — Paperclip — Running & Governing a Company of AI Agents: Docker-hosted company of agents, connecting OpenAI Codex, human approval gates, task tracking, budgets

		and audit (concepts + demo)
10:30–11:00	30 min	Hands-on: Lab 27: Setup Paperclip; Lab 28: Connect OpenAI Codex to Paperclip
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Hands-on: Lab 29: Configure Paperclip; Lab 30: Track AI Tasks; Lab 31: Automate AI Hiring; Lab 32: Setup AI Agents
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Hands-on: Lab 33: Security; Lab 34: Assign Task. Course recap and next steps
15:30–15:45	15 min	Tea break
15:45–16:15	30 min	Consolidation, Q&A and assessment preparation
16:15–16:30	15 min	Briefing for Assessment
16:30–17:30	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book
17:30–18:30	60 min	Practical Performance (PP) — hands-on agent-building tasks that follow the course labs, 1 hour, open book. PM digital attendance

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to course topics)

Topic / Platform	Weighting	Labs
Topic 01: Hermes Agent — Your Autonomous Personal Chief-of-Staff	33%	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5, Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13, Lab 14
Topic 02: OpenClaw — Automating a Business Back-Office	33%	Lab 15, Lab 16, Lab 17, Lab 18, Lab 19, Lab 20, Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26
Topic 03: Paperclip — Running & Governing a Company of AI Agents	34%	Lab 27, Lab 28, Lab 29, Lab 30, Lab 31, Lab 32, Lab 33, Lab 34